

Artavazd Maranjyan

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Current Position

2026–present **Postdoctoral Researcher and EPFL AI Center & Swiss AI Initiative Fellow**,
EPFL, Lausanne, Switzerland
Jointly hosted by **Volkan Cevher**'s LIONS group and **Martin Jaggi**'s MLO group.
Developing theoretically grounded methods to make large-scale AI training faster and more resource-efficient, with emphasis on distributed and asynchronous optimization, communication efficiency, and model parallelism.

Education

2023–2025 **Ph.D. in Computer Science**, *KAUST*, Thuwal, Saudi Arabia; **Advisor**: Peter Richtárik
Thesis: [First Provably Optimal Asynchronous SGD for Homogeneous and Heterogeneous Data](#)
2021–2023 **M.Sc. in Applied Statistics and Data Science**, *Yerevan State University*, Yerevan, Armenia
Thesis: [On local training methods](#); **Co-supervisors**: Peter Richtárik, Mher Safaryan
2017–2021 **B.Sc. in Informatics and Applied Mathematics**, *Yerevan State University*, Yerevan, Armenia
Thesis: [On the Convergence of Series in Classical Systems](#); **Supervisor**: Martin Grigoryan

Research Experience

03/2025 **Visiting Researcher, hosted by Yi-Shuai Niu**, *BIMSA*, Beijing, China
○ Collaborated on server-assisted federated learning.
○ Delivered invited seminars at Peking University (PKU), Beihang University (BUAA), and BIMSA.
04/23–08/23 **Researcher, Martin Grigoryan's group**, *Yerevan State University*, Yerevan, Armenia
○ Investigated universal functions for the Vilenkin and Haar systems across different function spaces.
06/22–01/23 **Research Intern, Peter Richtárik's group**, *KAUST*, Thuwal, Saudi Arabia
○ Contributed to GradSkip (TMLR), a communication-efficient local optimization method with improved complexity.
01/22–08/23 **Machine Learning Researcher**, *YerevaNN*, Yerevan, Armenia
○ Researched optimization methods for federated learning.

Industry Experience

07/21–06/22 **Co-Founder**, *OnePick*, Yerevan, Armenia
○ Built an MVP that helped content creators and marketers select photos matching their text descriptions.
○ Advanced through [FAST's InVent 2.0](#): team formation, ideation, product development, and pitching.
○ Secured initial funding and continued development beyond the program.
07/21–09/21 **Backend Developer**, *EXALT Technologies Ltd*, Yerevan, Armenia
○ Contributed to backend development for a Nutanix project.
06/21–07/21 **Machine Learning Research Engineer**, *FAST*, Yerevan, Armenia
○ Developed machine-learning models for fraud detection.
○ Built machine-learning and statistical models for data-driven forecasting.
09/19–01/21 **Software Engineer in Test**, *Picsart*, Yerevan, Armenia
○ Developed test-automation solutions across several mobile and web applications.
○ Collaborated across engineering, QA, product, and business teams to diagnose and resolve product issues.
○ Proposed improvements to DevOps processes and Jenkins CI pipelines.

Awards

06/2026 **EPFL AI Center and Swiss AI Initiative Postdoctoral Fellowship**, *EPFL*
Competitive fellowship supporting independent postdoctoral research (CHF 5,000 in annual travel funding)
05/2025 **CEMSE Dean's List Award**, *KAUST*
Awarded for excellent academic and research performance (USD 2,500)
09/2023 **Dean's Award**, *KAUST*
Merit-based award for selected incoming KAUST students (USD 18,000 over three years)
05/2021 **Outstanding Final Project Award**, *Yerevan State University*
Recognized for an outstanding bachelor's thesis (one of 6 recipients among 250+ students)

Publications

15. "LOSCAR-SGD: Local SGD with Communication-Computation Overlap and Delay-Corrected Sparse Model Averaging." Y. Maziane, A. Mahran, A. Maranjyan, P. Richtárik. [arXiv:2605.20866](#), 2026.
14. "Ringmaster LMO: Asynchronous Linear Minimization Oracle Momentum Method." A. Sadiev, A. Maranjyan, I. Ilin, P. Richtárik. [arXiv:2605.18174](#), 2026.
13. "Rescaled Asynchronous SGD: Optimal Distributed Optimization under Data and System Heterogeneity." A. Mahran, A. Maranjyan, P. Richtárik. [arXiv:2605.13434](#), 2026; [poster].
12. "Rennala MVR: Improved Time Complexity for Parallel Stochastic Optimization via Momentum-Based Variance Reduction." Z. Tovmasyan, A. Maranjyan, P. Richtárik. [arXiv:2605.08871](#), 2026.
11. "Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity." A. Maranjyan, P. Richtárik. *ICLR* 2026, [arXiv:2509.22860](#); [poster] [slides].
10. "BiCoLoR: Communication-Efficient Optimization with Bidirectional Compression and Local Training." L. Condat, A. Maranjyan, P. Richtárik. [arXiv:2601.12400](#), 2026.
9. "ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning." A. Maranjyan, E. M. Saad, P. Richtárik, F. Orabona. *ICML* 2025, [arXiv:2502.00775](#); [poster].
8. "Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity." A. Maranjyan, A. Tyurin, P. Richtárik. *ICML* 2025, [arXiv:2501.16168](#); [poster] [slides] [video].
7. "MindFlayer SGD: Efficient Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times." A. Maranjyan, O. S. Omar, P. Richtárik. *UAI* 2025; *OPT* 2024, **Oral (top 5%)**, [arXiv:2410.04285](#); [poster] [slides] [video].
6. "GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity." A. Maranjyan, M. Safaryan, P. Richtárik. *TMLR* 2025, [arXiv:2210.16402](#); [poster] [slides] [video].
5. "LoCoDL: Communication-Efficient Distributed Learning with Local Training and Compression." L. Condat, A. Maranjyan, P. Richtárik. *ICLR* 2025, **Spotlight (top 5.1%)**, [arXiv:2403.04348](#); [poster] [video].
4. "Differentially Private Random Block Coordinate Descent." A. Maranjyan, A. Sadiev, P. Richtárik. *OPT* 2024, [arXiv:2412.17054](#); [poster].
3. "Menshov-type theorem for divergence sets of sequences of localized operators." M. Grigoryan, A. Kamont, A. Maranjyan. *Journal of Contemporary Mathematical Analysis*, vol. 58, no. 2, pp. 81–92, 2023.
2. "On the divergence of Fourier series in the general Haar system." M. Grigoryan, A. Maranjyan. *Armenian Journal of Mathematics*, vol. 13, pp. 1–10, 2021.
1. "On the unconditional convergence of Faber-Schauder series in L^1 ." T. Grigoryan, A. Maranjyan. *Proceedings of the YSU A: Physical and Mathematical Sciences*, vol. 55, no. 1 (254), pp. 12–19, 2021.

Reviewing

TMLR (2024–present); NeurIPS and ICML (2026); ICLR (2025); SIMODS and JMLR (2024).

Selected Talks and Poster Presentations

- 26/04/2026 **Invited talk**, *Protocol Learning: Decentralized Collaborative Learning at Scale (ICLR 2026 Workshop)*, Rio de Janeiro, Brazil [video] [slides]
- 29/07/2025 **Talk and poster**, *35^o Colóquio Brasileiro de Matemática*, IMPA, Rio de Janeiro, Brazil [slides] [poster]
- 16/04/2025 **Invited talk**, *FLOW Talk #126, Federated Learning One World Seminar (FLOW)*, Online [video] [slides]
- 17/03/2025 **Invited talk**, *Academic Report of the School of Mathematical Sciences*, Beihang University (BUAA), Beijing, China, hosted by Jiaxin Xie [slides]
- 13/03/2025 **Invited talk**, *Optimization Seminar*, BIMS, Beijing, China, hosted by Yi-Shuai Niu [slides]
- 12/03/2025 **Invited talk**, *Seminar*, Peking University, Beijing, China, hosted by Kun Yuan [slides]
- 21/11/2024 **Invited talk**, *MLR Weekly Seminar, Machine Learning Research at Apple*, Online, hosted by Samy Bengio [slides]
- 13/03/2024 **Poster**, *The Machine Learning Summer School in Okinawa 2024*, OIST, Okinawa, Japan [poster]
- 06/10/2023 **Invited talk**, *Algorithms & Computationally Intensive Inference seminars*, University of Warwick, Coventry, England [slides]
- 07/12/2022 **Invited talk**, *FLOW Talk #88, Federated Learning One World Seminar (FLOW)*, Online [video]